

Assignment Operators =

Used to store or update values in variables.

Basic Assignment

```
x = 10
```

Output:

</> Python

  Run

Here:

- `x = 10` stores value 10 into x

Types of Assignment Operators

Operator	Meaning	Example	Same As
<code>=</code>	Assign	<code>x = 5</code>	Store 5
<code>+=</code>	Add and assign	<code>x += 3</code>	<code>x = x + 3</code>
<code>-=</code>	Subtract and assign	<code>x -= 3</code>	<code>x = x - 3</code>
<code>*=</code>	Multiply and assign	<code>x *= 3</code>	<code>x = x * 3</code>
<code>/=</code>	Divide and assign	<code>x /= 3</code>	<code>x = x / 3</code>
<code>//=</code>	Floor divide assign	<code>x //= 3</code>	<code>x = x // 3</code>
<code>%=</code>	Modulus assign	<code>x %= 3</code>	<code>x = x % 3</code>
<code>**=</code>	Power assign	<code>x **= 3</code>	<code>x = x ** 3</code>

Examples

+=

```
x = 10  
  
x += 5  
  
print(x)
```

Output:

 **Python** **Run****--**

```
x = 10  
  
x -= 3  
  
print(x)
```

Output:

 **Python** **Run**

***=**

```
x = 10  
  
x *= 2  
  
print(x)
```

Output:

</> Python



▶ Run

Comparison Operators

Used to **compare two values**.

Result is always:

- True
- False

Types of Comparison Operators

Operator	Meaning	Example	Result
==	Equal to	5 == 5	True
!=	Not equal	5 != 3	True
>	Greater than	5 > 3	True
<	Less than	5 < 3	False
>=	Greater or equal	5 >= 5	True
<=	Less or equal	5 <= 3	False

Examples

Equal Operator ==

```
a = 10
b = 10

print(a == b)
```

Output:

</> Python

  Run

Not Equal !=

```
a = 10  
b = 5  
  
print(a != b)
```

Output:

</> Python



▶ Run

Greater Than >

</> Python



▶ Run

Output:

</> Python



▶ Run

Less Than <

</> Python



▶ Run

Output:

</> Python



▶ Run

MOST IMPORTANT DIFFERENCE ★

Symbol

Meaning

=

Assign value

==

Compare values

Example

```
x = 5      # assigning  
  
print(x == 5)  # comparing
```

Output:

</> Python

 Run

Easy Memory Trick

- = → Put value into box 
- == → Ask question 
- += → Add and save
- > < → Compare bigger/smaller



list all

All Assignment Operators =

Operator	Example	Same As
=	x = 5	Assign value
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3

Operator	Example	Same As
<code>*=</code>	<code>x *= 3</code>	<code>x = x * 3</code>
<code>/=</code>	<code>x /= 3</code>	<code>x = x / 3</code>
<code>//=</code>	<code>x //= 3</code>	<code>x = x // 3</code>
<code>%=</code>	<code>x %= 3</code>	<code>x = x % 3</code>
<code>**=</code>	<code>x **= 3</code>	<code>x = x ** 3</code>
<code>&=</code>	<code>x &= 3</code>	<code>x = x & 3</code>
<code>'</code>	<code>= `</code>	<code>`x</code>
<code>^=</code>	<code>x ^= 3</code>	<code>x = x ^ 3</code>
<code>>>=</code>	<code>x >>= 3</code>	<code>x = x >> 3</code>
<code><<=</code>	<code>x <<= 3</code>	<code>x = x << 3</code>

All Comparison Operators

Operator	Meaning	Example
<code>==</code>	Equal to	<code>5 == 5</code>
<code>!=</code>	Not equal to	<code>5 != 3</code>

Operator	Meaning	Example
>	Greater than	5 > 3
<	Less than	5 < 3
>=	Greater than or equal	5 >= 5
<=	Less than or equal	5 <= 3

All Arithmetic Operators + -

Operator	Meaning	Example
+	Addition	5 + 2
-	Subtraction	5 - 2
*	Multiplication	5 * 2
/	Division	5 / 2
//	Floor Division	5 // 2
%	Modulus	5 % 2
**	Power	5 ** 2

All Logical Operators

Operator	Meaning
and	Both True
or	Any one True
not	Opposite

All Identity Operators

Operator	Meaning
is	Same object
is not	Different object

All Membership Operators

Operator	Meaning
in	Present
not in	Not present

All Bitwise Operators ⚡

Operator	Meaning
&	AND
^	XOR
~	NOT
<<	Left Shift
>>	Right Shift

Complete Operator Categories in Python 🔄

1. Arithmetic Operators
2. Assignment Operators
3. Comparison Operators
4. Logical Operators
5. Identity Operators
6. Membership Operators
7. Bitwise Operators



basic dsa program

Basic DSA Programs in Python

DSA = Data Structures and Algorithms

These are beginner-friendly important programs.

1. Array Traversal

```
arr = [10, 20, 30, 40, 50]

for i in arr:
    print(i)
```

Output:

```
10
20
30
40
50
```

2. Linear Search

Find element in array.

```
arr = [10, 20, 30, 40, 50]

key = 30

for i in arr:
    if i == key:
        print("Element Found")
        break
```

Output:

 Python



 Run

3. Find Largest Element

```
arr = [5, 10, 2, 8, 20]

largest = arr[0]

for i in arr:
    if i > largest:
        largest = i
```

```
print("Largest:", largest)
```

Output:

```
</> Python
```



▶ Run

4. Reverse an Array

```
arr = [1, 2, 3, 4, 5]

print(arr[::-1])
```

Output:

```
</> Python
```



▶ Run

5. Bubble Sort

```
arr = [5, 1, 4, 2, 8]
```

```
n = len(arr)

for i in range(n):
    for j in range(0, n-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

Output:

 Python



 Run

6. Stack Using List

LIFO = Last In First Out

```
stack = []

stack.append(10)
stack.append(20)
stack.append(30)

print(stack)

stack.pop()

print(stack)
```

Output:

```
[10, 20, 30]
```

```
10 20 30
```

```
Python
```

7. Queue Using List

FIFO = First In First Out

```
queue = []
```

```
queue.append(10)
```

```
queue.append(20)
```

```
queue.append(30)
```

```
print(queue)
```

```
queue.pop(0)
```

```
print(queue)
```

Output:

```
[10, 20, 30]
```

```
10 20 30
```

```
Python
```

8. Fibonacci Series

```
n = 10

a = 0
b = 1

for i in range(n):
    print(a)

    c = a + b
    a = b
    b = c
```

Output:

```
0
1
1
2
3
5
8
13
21
34
```

9. Factorial Program

```
n = 5

fact = 1

for i in range(1, n+1):
    fact = fact * i

print(fact)
```

Output:

 Python



 Run

10. Palindrome Check

```
n = "madam"

if n == n[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Output:

</> Python



▶ Run

11. Prime Number

```
n = 7

count = 0

for i in range(1, n+1):
    if n % i == 0:
        count += 1

if count == 2:
    print("Prime")
else:
    print("Not Prime")
```

Output:

</> Python



▶ Run

12. Binary Search 🔍

```
arr = [1, 2, 3, 4, 5, 6, 7]

key = 5

low = 0
high = len(arr) - 1

while low <= high:

    mid = (low + high) // 2

    if arr[mid] == key:
        print("Found")
        break

    elif arr[mid] < key:
        low = mid + 1

    else:
        high = mid - 1
```

Output:

 Python



 Run

MOST IMPORTANT DSA TOPICS

- Arrays

- Strings
- Linked List
- Stack
- Queue
- Searching
- Sorting
- Recursion
- Trees
- Graphs



give me ai tools which i can complete project within 10 min if i give prompt

If your goal is "**give a prompt → get a working project quickly**", these are the best AI tools in 2026:

1. Bolt.new (Best for Beginners)

- Create full-stack web apps from prompts
- Generates frontend + backend
- Can deploy directly
- Good for:
 - Employee Management System
 - Translator App
 - To-Do App

- Portfolio Website
- Expense Tracker

Example Prompt:

Create an Employee Management System using React, Node.js, and MySQL with login, CRUD operations, search, pagination, and responsive UI.

2. Lovable

- Build SaaS applications using prompts
- Beautiful UI automatically
- Database integration

Projects:

- CRM
 - HR Management
 - AI Chatbot
 - Sales Dashboard
-

3. Replit AI

- Creates and runs code instantly
- No setup required
- Good for Python projects

Projects:

- Cyberbullying Detection
- Flask Applications

- AI Translator
 - Student Management System
-

4. Cursor ↗

- AI-powered VS Code alternative
- Understands entire project
- Can create files automatically

Use for:

- Python
 - Django
 - Flask
 - Machine Learning
 - SQL Projects
-

5. GitHub Copilot ↗

- Works inside VS Code
 - Generates code from prompts
 - Excellent for learning
-

6. v0 by Vercel ↗

- Generates professional UI from prompts
- Best for frontend projects

Example:

Create a modern employee dashboard with sidebar, charts, employee cards, dark mode, and responsive design.

7. n8n

- AI Automation Tool
- No-code workflow builder

Projects:

- AI Sales Agent
- Email Automation
- Lead Generation
- WhatsApp Bot
- Resume Screening

Since you're learning Python, SQL, AI tools, and automation, I would focus on:

1. **Cursor** (coding)
2. **GitHub Copilot** (learning)
3. **Bolt.new** (quick projects)
4. **n8n** (automation)
5. **Lovable** (full SaaS apps)

10-Minute Project Ideas

- AI Translator
- Employee Management System
- Student Management System
- Expense Tracker
- AI Resume Analyzer

- AI Interview Bot
- Cyberbullying Detection System
- AI Sales Agent
- WhatsApp Automation Bot
- SQL Dashboard

For interviews, building **Employee Management System + AI Translator + Cyberbullying Detection + AI Sales Agent (n8n)** will give you a stronger portfolio than creating many small projects.



screenshot in laptop

If you want to take a **screenshot on a Windows laptop**, use these shortcuts:

Capture Entire Screen

PrtScn (Print Screen)

- Press `PrtScn`
- Screenshot is copied to clipboard
- Open Paint or Word → Press `Ctrl + V` → Save

Save Screenshot Automatically

Windows + PrtScn

- Press `Windows + PrtScn`

- Screen flashes briefly
- Screenshot is saved automatically in:

Pictures > Screenshots



Capture Selected Area

Windows + Shift + S

- Screen dims
- Select the area you want
- Screenshot is copied to clipboard
- Paste with `Ctrl + V`

Capture Active Window Only

Alt + PrtScn

- Captures only the current window
- Paste with `Ctrl + V`

Open Snipping Tool

- Press `Windows`
- Type **Snipping Tool**
- Open it and click **New**
- Select the area to capture
- Save the screenshot

Most Useful Shortcut ★

Windows + Shift + S



This is the shortcut most professionals use because you can capture only the required part of the screen.



For a **full-screen screenshot** on a Windows laptop:

Method 1: Save Automatically ✓

Press:

Windows + PrtScn



- Captures the entire screen.
- Automatically saves the image in:

This PC > Pictures > Screenshots



Method 2: Copy to Clipboard

Press:

PrtScn



- Captures the entire screen.
- Doesn't save automatically.
- Open Paint, Word, WhatsApp, etc., and press:

Ctrl + V



to paste it.

If PrtScn doesn't work

On some laptops, use:

Fn + PrtScn



or

Fn + Windows + PrtScn



depending on your keyboard model.



if i have to full screen

If you want to capture the **entire screen (full screen)**:

On Windows Laptop

Press:

Windows + PrtScn



- ☒ Captures the whole screen
- ☒ Saves automatically

Find it here:

This PC → Pictures → Screenshots



If that doesn't work, try:

Fn + Windows + PrtScn



because some laptops require the **Fn** key.

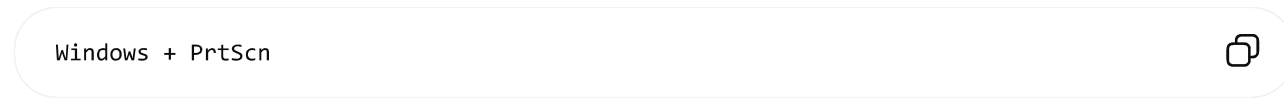
If you mean **making an application/browser full screen** before taking the screenshot, press:

F11



This makes Chrome, Edge, VS Code, etc. go full screen.

Then press:



to capture the entire screen.



i have laptop i want to get print what are the steps

If you want to **print a document from your laptop**, follow these steps:

Step 1: Connect the Printer

- Turn on the printer.
- Connect it to your laptop:
 - **USB cable** (easiest), or
 - **Wi-Fi** (both laptop and printer on the same network).

Step 2: Open the File

Open the document, PDF, image, or webpage you want to print.

Step 3: Open Print

Press:

Ctrl + P



This opens the Print window in most applications.

Step 4: Select Your Printer

Choose your printer from the printer list.

Step 5: Choose Print Settings

- Number of copies
- Color or black & white
- Page range (all pages or specific pages)
- Paper size

Step 6: Click Print

Click the **Print** button.

If Your Printer Doesn't Appear

1. Open **Settings** → **Bluetooth & devices** → **Printers & scanners**.
2. Click **Add device**.
3. Select your printer and install it.

If You Don't Have a Printer

- Save the file to a USB drive or your phone.
- Take it to a nearby print shop and ask them to print it.

What exactly do you want to print—a Word document, PDF, image, or webpage? I can give more specific steps.

